

Computational Modeling and Physics-Guided Surrogate Prediction of Euler-Bernoulli Beam Response

Independent Computational Project | MATLAB | Graduate Application Portfolio | July 2026

FEM VALIDATION 0.000011% Deflection relative L2 error	SURROGATE ACCURACY > 0.9995 Held-out test R2 for all outputs	UNSEEN GENERALIZATION 1.086% Maximum relative L2 error
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Figure 1. Simply supported Euler-Bernoulli beam under a full-span distributed load and an off-center point load.

PROJECT OBJECTIVE

- Build and verify a two-node Euler-Bernoulli finite-element solver.
- Generate parametric response fields for distributed and point loading.
- Develop accurate neural-network surrogates without violating known support conditions or the shear discontinuity.

CORE CONTRIBUTIONS

- Analytical validation, force equilibrium checks, and mesh-convergence analysis.
- Case-level training/validation/test split across 350 loading cases.
- Hard boundary-condition transformations and piecewise shear networks.

Scope: fixed geometry, material, and simply supported boundary conditions; load family consists of one full-span UDL and one point load.

1. Finite-Element Model, Analytical Validation, and Mesh Convergence

Model item	Specification	Reference value
Beam	Prismatic, simply supported	$L = 6.0\text{ m}$
Material	Linear elastic	$E = 200\text{ GPa}$
Section	Rectangular: $b = 0.15\text{ m}$, $h = 0.30\text{ m}$	$I = 3.375 \times 10^{-4}\text{ m}^4$
Discretization	Two-node cubic Hermite beam elements	40 elements / 41 nodes
Loading	Full-span UDL + one point load	$q = 8\text{ kN/m}$, $P = 25\text{ kN}$, $a/L = 0.38$

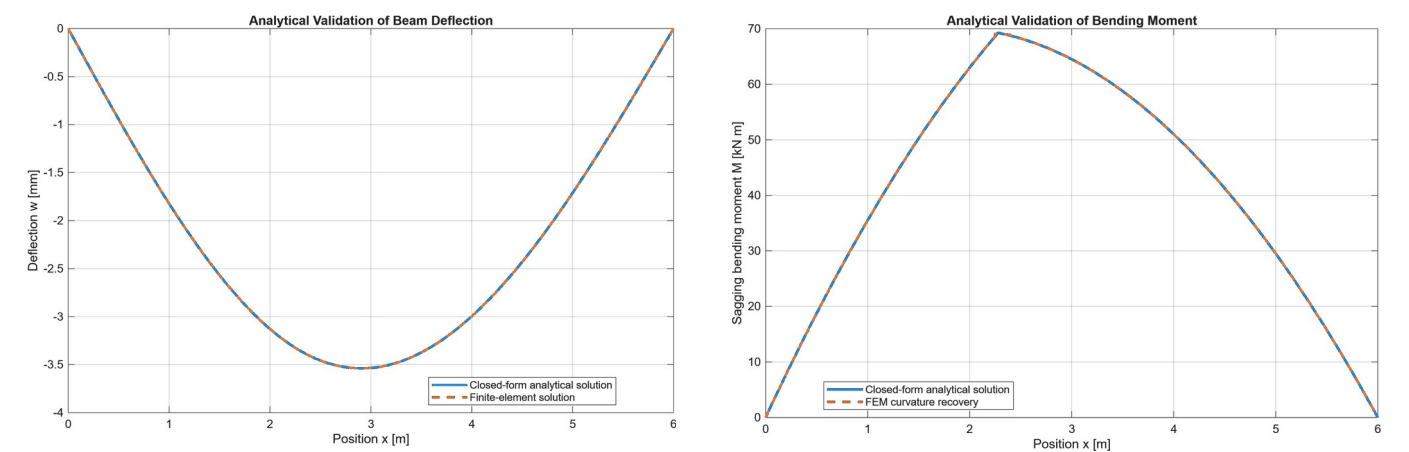


Figure 2. Closed-form solutions and FEM results are visually coincident for deflection and bending moment.

Verification metric	FEM result	Reference / error
Left support reaction	39.500 kN	Exact: 39.500 kN
Right support reaction	33.500 kN	Exact: 33.500 kN
Maximum deflection	3.538329 mm	Exact: 3.538329 mm
Deflection relative L2 error	-	0.000011%
Moment relative L2 error	-	0.052340%
Vertical equilibrium residual	-	$6.71 \times 10^{-8}\text{ N}$

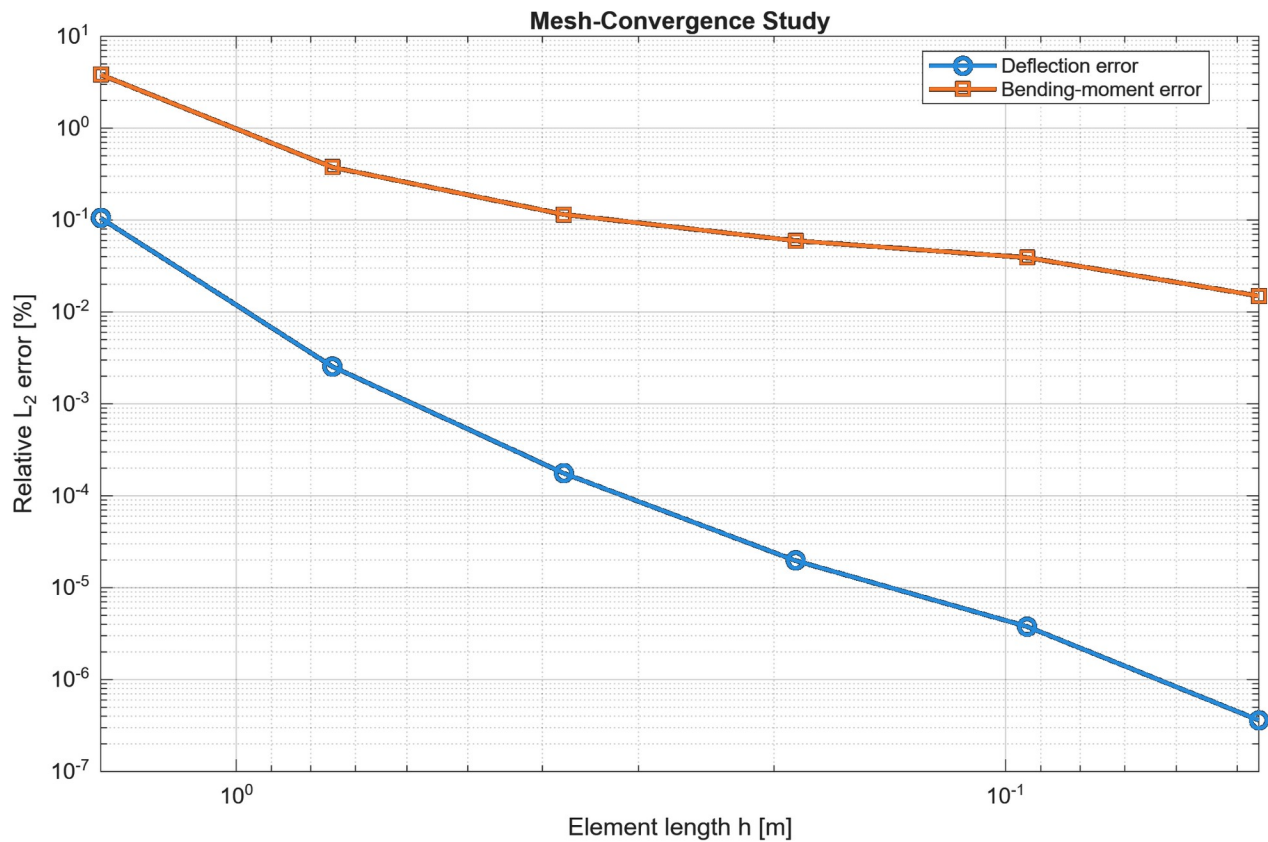


Figure 3. Mesh refinement from 4 to 128 elements produces systematic reductions in displacement and bending-moment errors.

Interpretation. The displacement error decreased from 0.1048% to $3.62 \times 10^{-7}\%$, while the bending-moment error decreased from 3.8218% to 0.01496%. The slower moment convergence is consistent with recovering a second-derivative quantity from the displacement field.

2. Physics-Guided, Discontinuity-Aware Surrogate Architecture

Computational workflow and surrogate architecture



Figure 4. Verified FEM data generation, case-level splitting, and four response-specific surrogate networks.

PHYSICS-GUIDED OUTPUT TRANSFORMS

$w_{\text{hat}}(x) = \xi(1 - \xi) N_w(\xi, q, P, a)$
 $M_{\text{hat}}(x) = \xi(1 - \xi) N_M(\xi, q, P, a)$
 $\xi = x/L$; therefore $w_{\text{hat}}(0) = w_{\text{hat}}(L) = 0$ and $M_{\text{hat}}(0) = M_{\text{hat}}(L) = 0$ exactly.

Shear treatment: independent left- and right-branch networks avoid forcing a continuous approximation across the point-load jump.

DATASET AND TRAINING DESIGN

- 350 complete loading cases; 41 spatial nodes per case.
- $q = 2\text{--}15 \text{ kN/m}$; $P = 5\text{--}45 \text{ kN}$; $a/L = 0.15\text{--}0.85$.
- 245 training, 53 validation, and 52 held-out test cases.
- Features: ξ , normalized q and P , $\alpha = a/L$, $\xi\text{-}\alpha$, and $|\xi\text{-}\alpha|$.
- Scaled conjugate-gradient training with validation-based early stopping.

Held-out test performance

Response	RMSE	Test R2
Deflection	0.013080 mm	0.999915
Bending moment	0.552811 kN m	0.999559
Shear force	0.099024 kN	0.999987

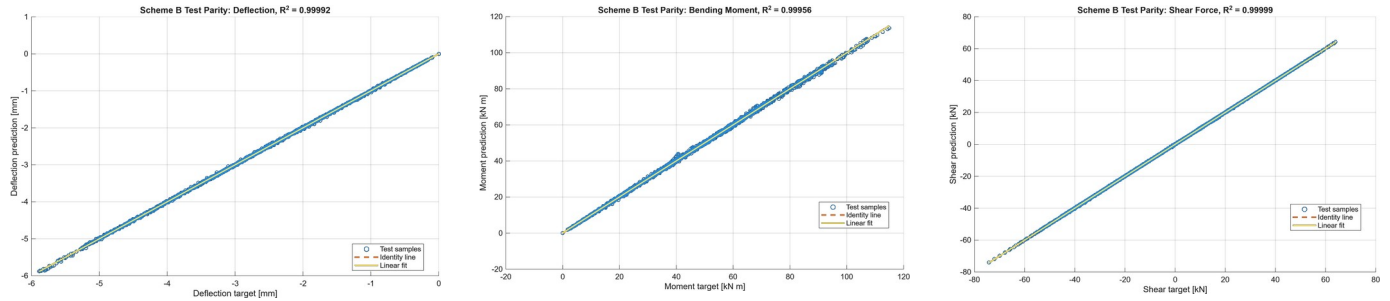


Figure 5. Per-output parity plots for held-out load cases; all three response fields closely follow the identity line.

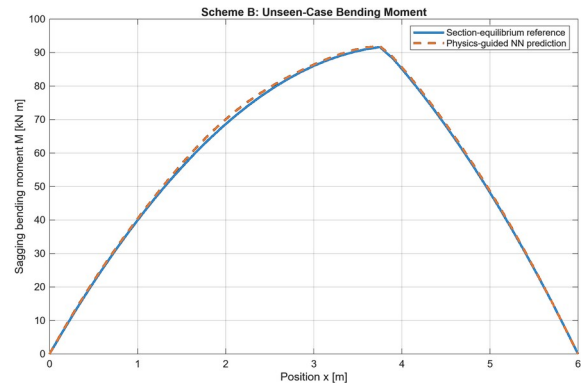
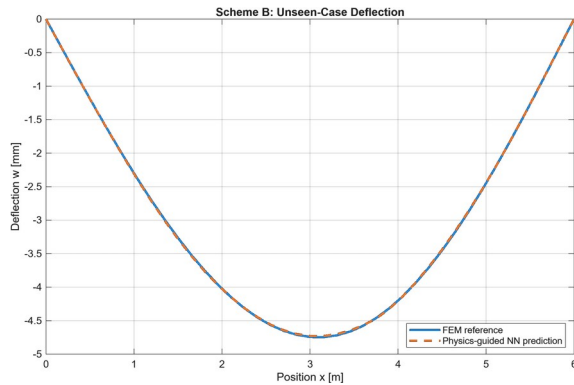
Methodological note: this is a physics-guided data-driven surrogate, not a PINN; no governing-equation residual is included in the loss function.

Improvement over the initial multi-output baseline

Response	Baseline test R2	Scheme B test R2	Unseen L2 error
Deflection	0.6990	0.9999	12.73% -> 0.247%
Bending moment	0.9852	0.9996	6.64% -> 1.09%
Shear force	0.9887	1.0000	10.52% -> 0.182%

3. Reserved Unseen-Case Evaluation and Engineering Interpretation

Reserved case: $q = 11.3 \text{ kN/m}$, $P = 31.7 \text{ kN}$, $a/L = 0.630$



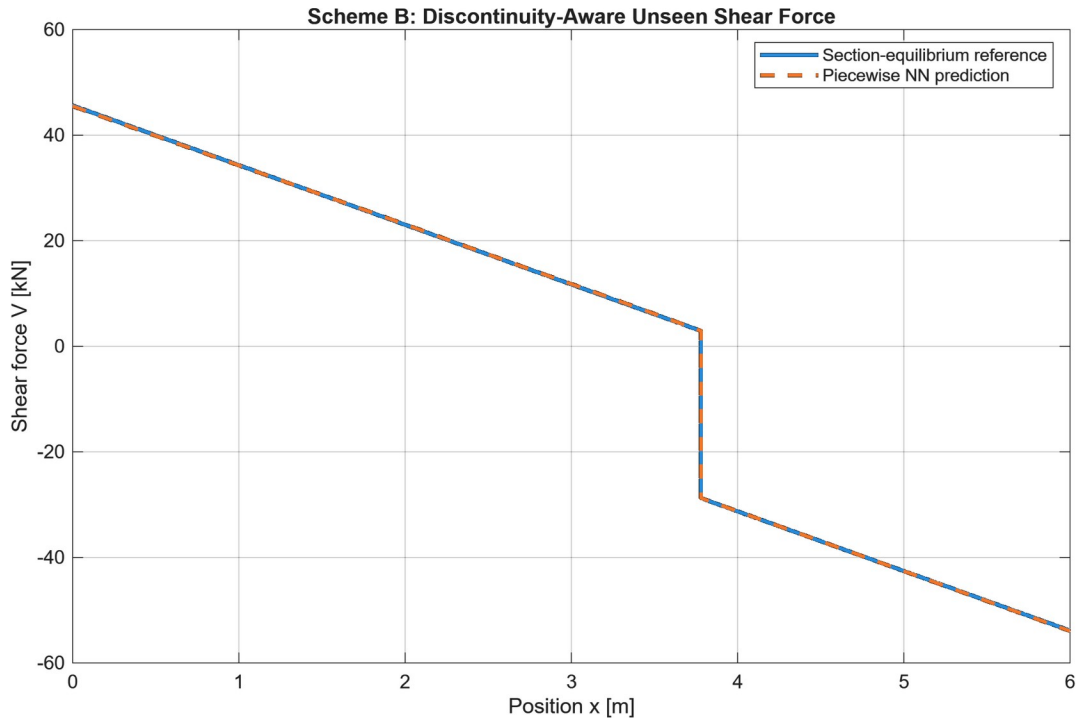


Figure 6. Scheme B reproduces the unseen deflection and moment fields and preserves the point-load shear discontinuity.

Response	RMSE	R2	Relative L2 error
Deflection	0.008184 mm	0.999971	0.246782%
Bending moment	0.668925 kN m	0.999443	1.086163%
Shear force	0.061537 kN	0.999997	0.181741%

PHYSICAL CONSISTENCY • Support deflection residual: 0 mm • Support moment residual: 0 kN m • Predicted shear jump: -31.6735 kN • Exact shear jump: -31.7000 kN • Shear-jump error: 0.0836%	SCOPE AND LIMITATIONS • Fixed L, E, I, cross-section, and simply supported boundary condition. • Interpolation within the stated q, P, and a/L ranges; extrapolation was not established. • Synthetic deterministic FEM data; no experimental measurement noise. • One UDL plus one point load; additional load and support types remain future work.
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Conclusion. The project demonstrates a verified computational-mechanics workflow and a carefully structured surrogate model that incorporates known boundary and discontinuity behavior. Within the defined parameter domain, the final architecture achieved held-out R2 values above 0.9995 and unseen-case relative L2 errors below 1.1%.